

Be Tuan Minh

Senior Backend Engineer

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Professional Summary

Backend Engineer with 5+ years of experience delivering .NET services for large-scale education and assessment platforms. Owns backend work from system design and data modeling through implementation, migration, production rollout, and troubleshooting. Current scope includes IOE (35M+ accounts; peak production load up to 200K concurrent users), centralized SSO, AI-assisted learning, ranking systems, and legacy modernization. Strong in C#, ASP.NET Core, SQL, Redis, messaging, OAuth 2.0/OIDC, containers, and observability.

Experience

Backend Engineer, VTC Online

Mar 2026 – Present

National-scale Education Platforms & Identity

- Deliver backend capabilities across IOE, IOK, IOSTEM, and IOM; IOE serves **35M+ registered accounts** and reaches peak production loads of up to **200K concurrent users**.
- Implemented centralized SSO across the product ecosystem using **OAuth 2.0/OIDC, PKCE**, secure token lifecycles, and cross-application session flows.
- Delivered IELTS Mastery and AI Writing workflows with skill-level mastery tracking, AI-assisted scoring, and actionable writing feedback.
- Built ranking capabilities for IOE and IOM, covering score aggregation, deterministic ordering, and high-volume leaderboard delivery.
- Modernizing IOE v3 by migrating legacy ASP.NET MVC functionality to React + .NET while preserving business rules and production continuity.
- Developed backend integrations for IOE Speak Out 2026 and strengthened reliability through Redis caching, asynchronous processing, SQL tuning, observability, and release controls.

Software Engineer, AvePoint Vietnam

Jan 2023 – Mar 2026

IAS 2.0 — Technical Education Platform

- Implemented CQRS and event-driven workflows with RabbitMQ to coordinate services using eventual consistency.
- Built .NET microservices backed by PostgreSQL read/write replication and MongoDB sharding.
- Reduced inter-service latency by **35%** using Redis caching, asynchronous messaging, and integration-flow optimization.
- Operated Kubernetes workloads with Grafana and Prometheus, improving production monitoring and incident diagnosis.

Examina — Online Assessment & Proctoring

- Built .NET backend services for online assessment and real-time WebRTC proctoring with MediaSoup, supporting **3,000+ concurrent students**.
- Designed asynchronous cheating-detection signals and high-throughput audit logging for exam events.
- Reduced critical API latency from **600 ms to 280 ms** through SQL query/index tuning and Redis caching.
- Delivered containerized services using Docker, Kubernetes, Helm, Terraform, GitLab CI/CD, and Argo CD.

Skills

Backend: C#, .NET, ASP.NET Core, REST APIs, Dapper, EF Core, SQL

Data: SQL Server, PostgreSQL, MongoDB, Redis; query/index tuning, replication, sharding

Architecture: Microservices, DDD, CQRS, event-driven systems, Clean Architecture, BFF, API Gateway

Messaging & Reliability: RabbitMQ, MassTransit, Outbox, Saga, distributed locking, caching

Identity & Security: SSO, OAuth 2.0, OpenID Connect, PKCE, JWT, RBAC

Cloud & Delivery: Docker, Kubernetes, Helm, Terraform, AWS, GCP, GitLab CI/CD, GitHub Actions, Argo CD

Observability & Quality: Grafana, Prometheus, Elasticsearch, Loki, Sentry, xUnit, NUnit, Jest

Frontend Collaboration: React, TypeScript, JavaScript, Next.js

Education & Awards

FPT University, B.Sc. in Software Engineering

GPA: 3.3/4.0 | 30% Scholarship | Academic Excellence: Fall 2022 | Honors: Spring 2023, Fall 2024